Math 510 **Fall 2026**

Homework 2 **Drew Armstrong**

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1. Let V be a real inner product space. A set $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\} \subseteq V$ is called *orthogonal* if

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0 \text{ for all } i \neq j.$$

(a) Prove that every orthogonal set is an independent set.

(b) An orthogonal set $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is called *orthonormal* if we also have $\langle \mathbf{u}_i, \mathbf{u}_i \rangle = 1$ for all i. Let S be an orthonormal set and suppose that

$$\mathbf{v} = a_1\mathbf{u}_1 + a_2\mathbf{u}_2 + \ldots + a_k\mathbf{u}_k.$$

In this case, prove that $\langle \mathbf{v}, \mathbf{u}_i \rangle = a_i$ for all i, and

$$\|\mathbf{v}\|^2 = a_1^2 + a_2^2 + \cdots + a_k^2.$$

2. Let W be a vector space and let $U, V \subseteq W$ be subspaces.

(a) Prove that $U \cap V$ is a subspace, but $U \cup V$ need not be a subspace.

(b) Define the sum of subspaces by

$$U + V = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in U, \mathbf{v} \in V\}.$$

Prove that $U + V$ is a subspace of W.

(c) Suppose that $U \cap V = \{\mathbf{0}\}$. Show that for all vectors $\mathbf{w} \in U + V$ there exists a **unique** expression $\mathbf{w} = \mathbf{u} + \mathbf{v}$ with $\mathbf{u} \in U$ and $\mathbf{v} \in V$. [Hint: Suppose that $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$ with $\mathbf{u}, \mathbf{u}' \in U$ and $\mathbf{v}, \mathbf{v}' \in V$. Then ...]

(d) If $W = U + V$ and $U \cap V = \{\mathbf{0}\}$ then we say that W is the *direct sum* of U and V, and we write $W = U \oplus V$. [Hint: Let $\mathbf{u}_1, \ldots, \mathbf{u}_m$ be a basis of U and let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be a basis of V. Prove that $\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{v}_1, \ldots, \mathbf{v}_n$ is a basis of W.]

3. Let V be a real inner product space and let $U \subseteq V$ be a subspace. We define the *orthogonal complement* of U as follows:

$$U^\perp := \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

(a) Prove that $U^\perp$ is a subspace of V.

(b) Prove that $U \cap U^\perp = \{\mathbf{0}\}$.

(c) Prove that $V = U + U^\perp$, and hence $V = U \oplus U^\perp$. [Hint: Suppose that $\dim U = k$ and $\dim V = n$. From the Gram-Schmidt process (see below), it is possible to find an orthonormal basis $\mathbf{u}_1, \ldots, \mathbf{u}_n$ for V such that $\mathbf{u}_1, \ldots, \mathbf{u}_k$ is a basis for U. Show that $\mathbf{u}_{k+1}, \ldots, \mathbf{u}_n$ are in $U^\perp$ and use this to show that every $\mathbf{v} \in U$ can be expressed as $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$.]

Future problems will involve projection, Gram-Schmidt, possibly Fourier series.